

Question and Answers.

1. Find $L\{f(t)\}$ if $f(t) = \begin{cases} \sin 2t, & 0 < t < \pi \\ 0, & t > \pi \end{cases}$

Solution: $L\{f(t)\} = \int_0^{\pi} e^{-st} \sin 2t \, dt$

$$= \left[L\{f(t)\} = \frac{e^{-st}}{s^2 + 4} (-s \sin 2t - 2 \cos 2t) \right]_{t=0}^{\pi}$$

$$= \frac{-1}{s^2 + 4} [2 e^{-s\pi} - 2]$$

$$= \frac{2}{s^2 + 4} (1 - e^{-\pi s}).$$

2 . Find LT of $\sin 5t \cos 2t$

Solution: $f(t) \frac{1}{2} \{\sin 7t + \sin 3t\}$

$$L[f(t)] = \frac{1}{2} \left[\frac{7}{s^2 + 49} + \frac{3}{s^2 + 9} \right]$$

3 . Evaluate $\int_0^{\infty} \frac{\cos 6t - \cos 4t}{t} \, dt$ using LT.

Solution: WKT $\int_0^{\infty} \frac{\cos 6t - \cos 4t}{t} \, dt = L \left[\frac{\cos 6t - \cos 4t}{t} \right]$

$$L \left[\frac{\cos 6t - \cos 4t}{t} \right] = \frac{1}{2} [\log (s^2 + 16) - \log (s^2 + 36)]_s^{\infty}$$

$$\int_0^{\infty} \frac{\cos 6t - \cos 4t}{t} \, dt = \frac{1}{2} \log \left(\frac{s^2 + 16}{s^2 + 36} \right)$$

Substituting $S = 0$ We get,

$$\int_0^{\infty} \frac{\cos 6t - \cos 4t}{t} dt = \log(2/3).$$

4 . Find LT of $\int_0^{\infty} \frac{e^{-t} \sin t}{t} dt$.

Solution: $L \left[\frac{e^{-t} \sin t}{t} \right] = \int_s^{\infty} L(e^{-t} \sin t) ds$

$$= \int_s^{\infty} \frac{1}{(s+1)^2 + 1} ds$$

$$L \left[\frac{e^{-t} \sin t}{t} \right] = [\tan^{-1}(s + 1)]_s^{\infty}$$

$$= \cot^{-1}(s + 1)$$

$$\int_0^{\infty} e^{-st} \cdot t \frac{e^{-t} \sin t}{t} dt = \cot^{-1}(s + 1)$$

Substituting $s=0$

$$\int_0^{\infty} e^{-st} \cdot \frac{e^{-t} \sin t}{t} dt = \cot^{-1}(1) = \frac{\pi}{4}$$

5. Find LT of the triangular wave of period $2a$ given by

$$f(t) = \begin{cases} t, & 0 < t < a \\ 2a - t, & a < t < 2a \end{cases}$$

Solution: $L\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^{2a} e^{-st} f(t) dt$

$$= \frac{1}{1 - e^{-2as}} \left\{ \int_0^a t e^{-st} f(t) dt + \int_a^{2a} (2a - t) e^{-st} dt \right\}$$

Applying Bernoulli's rule to each of the integrals, we get

$$\mathcal{L}\{f(t)\} = \frac{1}{s^2(1-e^{-2as})} (-e^{-sa} + 1 + e^{-2as} - e^{-as})$$

$$= \frac{(1-e^{-as})(1+e^{-as})}{s^2(1-e^{-as})(1+e^{-as})}$$

$$\mathcal{L}\{f(t)\} = \frac{1}{s^2} \tanh\left(\frac{as}{2}\right).$$